ORIGIN OF LIFE

* life emerged 3.8×10^9 years ago. (approx 750 million years after Earth was formed).

Types of organisms

Prokaryotes → Eukaryotes

* Prokaryotes → It is a single celled organism that lacks a membrane-bound nucleus (Karyon), mitochondria or any other membrane-bound organelle.

* Eukaryotes :- It is any organism whose cells contain a nucleus and other organelles enclosed within membranes.

* Various body systems present in a human body :-

1) Respiratory system
2) Cardiovascular system (heart and circulatory system)
3) Digestive system
4) Nervous system
5) Lymphatic system (or immune system)
6) Endocrine system
7) Integumentary system (or Exocrine system)
8) ~~Muscle~~ Musculoskeletal system
9) Urinary system
10) Reproductive system.

* 79 organs have been identified in the human body.

* latest organ discovered is tubarial salivary glands and are found behind the nose, in the nook where the nasal cavity meets the throat.

* skin is the largest and longest organ acc. to its size and weight.

* Pineal gland near the center of the brain, in a groove between the hemispheres is the smallest organ.

* Liver is the largest internal organ of the body.

* The heart is the first major functional organ to develop and starts to beat and pump blood at about 3 weeks into embryogenesis.

* The last major organ to develop and mature before birth are the lungs.

* The brain is the last to reach its final mature form and develops even after birth.

* sensory organs:-
1) eyes
2) ears
3) nose

4) skin
5) tongue

Mesentery : Newest organ discovered (79th organ)

It's work :-

→ Large single continuous stretch of tissue that support and position all the digestive organs in the abdomen.

→ The organ is responsible for transporting blood and lymphatic fluid between the intestine and the rest of the body.

* Types of tissues :-

1) Epithelial tissue
2) Connective tissue
3) Muscle tissue (3 types — skeletal (voluntary), smooth muscles, cardiac muscles)
4) Nervous tissue

* Total 37.2 trillion (10^{12}) cells in our body

* Total 200-220 cell types in the body.

* ~~Cell mem~~ Various organelles in a human cell :-

1) Cell membrane :- consists of double phospholipid layer and monolayer of protein scattered around phospholipid layer.

NOTE → We can infer that while cytosol is the fluid contained in the cell cytoplasm, cytoplasm is the entire contenst within the cell membrane.

2) Cytosol
3) ~~Cell~~ Cytoskeleton

4) Nucleus :- It consists of 3 main regions :-
1) Nuclear membrane
2) Nucleolus
3) chromatin

5) Nucleolus → functions as site of ~~protein~~ ribosome production , ribosomes then migrate to the cytoplasm through nuclear pores)

6) Ribosome → sites of protein synthesis in the cell.

7) Rough endoplasmic reticulum — (rough coz it carries ribosomes that represents sites of protein synthesis)

8) Smooth endoplasmic reticulum - function in cholesterol synthesis and breakdown, fat metabolism and detoxification of drugs.

9) Golgi apparatus :- (modifies and packages proteins, secrete vesicles, plasma membrane components and lysosomes).

10) Mitochondrion

1) Lysosomes - suicidal bag of cell; contain enzymes that digest non-usable materials within the cell.

2) Peroxisomes - detoxify harmful substances and break down free radicals; involved in catabolism of very long chain fatty acids.

3) Centrioles - involved in the organization of mitotic spindle fibers during cell division and in the completion of cytokinesis.

4) Centrosomes - one of the primary functions of centrosomes is to serve as the main organizing center for microtubules in the cell.

5) Vesicles :- involved in metabolism, transport, buoyancy control, temporary storage of food and enzymes.

* Macromolecules :-
1) Nucleic Acid
2) Carbohydrate
3) Lipid
4) Protein

NOTE → The body of a 57 Kg teenager would consist of about 1.2×10^{27} molecules.

→ The human body consists of about 2×10^{25} molecules, and more than 99% of them are water.

* There are approx. 7×10^{27} atoms in the avg human body. (70 Kg adult male)

* On an avg there are around 10^{14} atoms in a typical human cell.

* Ribosomes was first organelle formed by infolding of membrane (It can be wrong!)

* 6.4 billion bases in nuclear genome.

* Cellular genome more than nuclear genome (because mitochondria also has DNA)

* CCR5 present in cell membrane through which virus enter our cell.
(It is HIV Co-receptor).

* Virus can have both single stranded and double stranded DNA and RNA.

Imp Genetic material of CORONA VIRUS

* Archea is found in extreme condition

* Till date, Archea is ~~cond~~ considered non-pathogenic

* Due to its role in HIV infection, CCR5 has garnered significant attention in medical research

Some individuals carry a genetic mutation known as CCR5 - Δ32, which results in the deletion of a portion of the CCR5 gene.

Corona virus including the Severe Acute Respiratory syndrome coronavirus 2 (SARS-COV-2) responsible for COVID -19, are enveloped viruses :-

1) Single stranded RNA :- This RNA genome serves as the ~~blue~~ genetic bluprint for the virus and contains all the necessary instructions for viral replication and assembly.

2) Positive sense RNA :- The RNA genome of coronavirus is positive sense, meaning that it can be directly translated by host cell machinery to produce viral proteins.

This is in contrast to negative sense RNA viruses, which require the synthesis of complementary RNA strands before protein translation can occur.

3) Large Genome : The RNA genome of coronavirus is relatively large compared to many other RNA viruses, consisting of approximately 30, or to 40 or nucleotides.

4) Genetic variation

Characteristic	Prokaryote	Eukaryote
Nucleus	Absent	Present
Diameter of a typical cell	$\approx 1\ \mu m$	$10 - 100\ \mu m$
Cytoskeleton	Absent	Present
Cytoplasmic organelles	Absent	Present
DNA content (base pairs)	1×10^6 to 5×10^6	1.5×10^7 to 5×10^9
Chromosomes	Single circular DNA molecule	Multiple linear DNA molecules

NOTE

* Prokaryotic cells :-

→ higher surface area to volume ratio (SA : V)

→ higher metabolic rate

→ shorter generation time.

* Commonalities between Pro and Eukaryotes :-

1) Both have DNA
2) Both have cell membrane
3) Both have ribosomes.

EVOLUTION OF LIFE

Antibiotic affect bacteria only.

cells

~~Eukaryote~~ Eukaryotes — Prokaryotic

Eukaryotes: Animals, Protists, Plants

Prokaryotic: Archea, Bacteria

The same basic molecular mechanisms govern the lives of both prokaryotes and eukaryotes, indicating that all present day cells are descended from a single primordial ancestor.

* Timescale of evolution :-

4.6 billion years ago (Formation of Earth) → First cells (4 billion years ago) → First eukaryotes, Oxidative metabolism, Photosynthesis (3 Billion years ago) → 2 billion years ago (Multicellular organisms)

* Endosymbiosis theory :- It is a scientific hypothesis that proposes the origin of eukaryotic cells through a symbiotic relationship between different prokaryotic organisms.
This theory suggests that certain organelles within eukaryotic cells, such as mitochondria and chloroplasts, originated from free-living prokaryotic cells that were engulfed by ancestral eukaryotic cells in a mutualy beneficial symbiotic relationship.

Supporting evidence :- several lines of evidence support the endosymbiosis theory, including similarities in the structure and function of mitochondria and chloroplasts to free living bacteria, the presence of their own DNA (circular like bacterial DNA
they also replicate through binary fission.

* Evolution is the change in the inherited characteristics of biological populations over successive generations.

* origin of life on earth is also called Protobiogenesis.

* Theories of evolution :-

1) Theory of Special Creation :-
All living organisms that we see today were/are created by God.

2) Theory of Spontaneous Generation :-

→ States that life can originate from non living matter.
→ Aristotle proposed this theory.
→ Francesco Redi showed that maggots arise from the eggs of flies rather than directly from rotting
→ Later, Louis Pasteur did experiments with flasks with twisted necks (swan-neck flasks) and proved sterilised broths in swan neck flask remained sterile.
→ Unless microbes are introducing from outside from air, the broths remained sterile, and there was no microbial growth of microorganisms.

→ Pasteur's experiments disproved spontaneous generation theory by proving "life only comes from life."

3) Theory of chemical evolution :-

→ In 1920s British scientist J.B.S Haldane and Russian biochemist Aleksander Oparin independently set forth similar ideas concerning the conditions required for the origin of life on earth.

Abiogenesis

simple molecules
↓
Molecules of life
↓
macromolecules
↓
Polymer complexes
↓
Metabolic Networks
↓
Living cells.

* They suggested that under the strong reducing conditions in the atmosphere of early earth, inorganic molecules would spontaneously form organic molecules (simple sugars and amino acids).

* The spontaneous formation was shown by Stanley Miller by discharging electric sparks into a mixture of H_2, CH_4 and NH_3 in the presence of H_2O led to a formation of a variety of organic molecules including several amino acids.

NOTE → Heating dry mixtures of amino acids, for example results in their polymerisation to form polypeptides.

* This theory is accepted.

$CO_2 \rightarrow CO + [O]$

$CH_4 + 2[O] \rightarrow CH_2O + H_2O$

$CO + NH_3 \rightarrow HCN + H_2O$

$CH_4 + NH_3 \rightarrow HCN + 3H_2$ (~~[illegible]~~ BMA process)

The formaldehyde, ammonia, and HCN react by Strecker synthesis to form amino acids and other biomolecules.

$CH_2O + HCN + NH_3 \longrightarrow NH_2-CH_2-CN + H_2O$

$NH_2-CH_2-CN + 2H_2O \rightarrow NH_3 + NH_2-CH_2-COOH$
(glycine)

ii) Theory of Panspermia or Cosmozoic theory →

→ "cosmozoa" or "pansperms" (seeds everywhere) which were preserved inside the meteorites.

→ These meteorites might have evolved into present day life forms.

NOTE → Read Radio active dating from links (Done).

CRISPER BOOK

* Generally > 6.4 billion bases ~~pairs~~ in human genome
  (size of Y > size of X)

* **LUCA** → last Universal Common Ancestor. (~~Archea~~)

* In darwin's word :- inherited variation
  In today → genetic mutation

READ

* Mechanism of Antibiotic.

* **HUMAN GENOME :-**

→ It is diploid. It has 46 chromosomes means 23 pairs.

→ Human genome contains about 3.2 billion base pairs of DNA. However, within the genome there are only about 20,000 genes that code for proteins, ~~compli~~ comprising only 1.5% of the genome.

EMPIRE

- Cellular
  - Bacterya
  - Archea
  - Eukarya
- Viral

* Giant viruses are so huge they can even have their own parasites - the biggest are

almost the same size as some bacteria

→ virus typically encodes some proteins essential for viral replication, but they never contain the full set of genes for the proteins and RNAs required for translation, membrane function or metabolism.

- The human genome is made up of 20,000 instructions called genes.
But add all the genes in our microbiome together and the figure comes out between 2 and 20 million microbial genes.

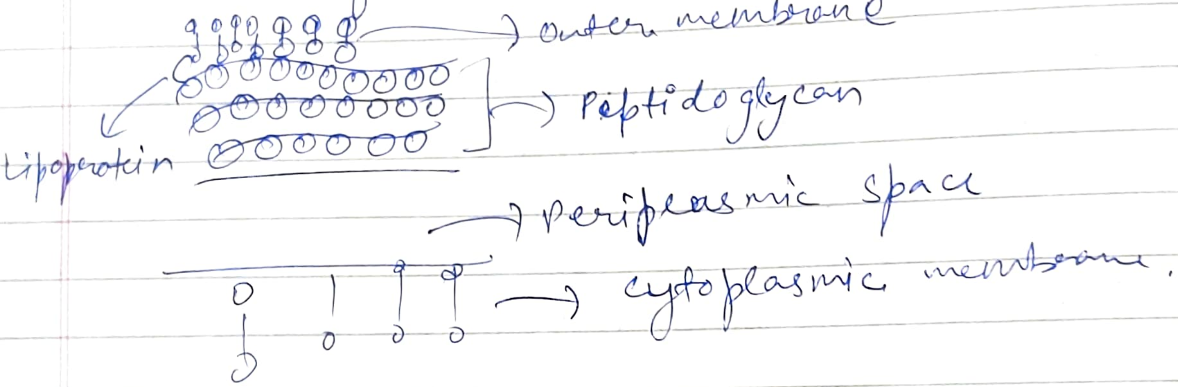
+ Oxygen holocaust → when a lot of O_2 was ~~breath~~ released by the prokaryotes and the other ~~sig~~ single celled organisms ~~that~~ died out of it.

+ CRISPR → Clustered Regularly Interspaced Short Palindromic Repeats.
→ It refers to the family of DNA sequences found within the genomes of bacteria and other microorganisms.
CRISPR sequences are part of the bacterial immune system and provide a defense mechanism against foreign genetic elements, such as viruses and plasmids. (gene editing basically).

* 3 domains of life

Bacteria | ~~Archea~~ Archaea | Eukarya

Structure of Bacteria :-



* Antibiotic target the central processes of bacteria :-
  1) RNA synthesis (attacks the 30S Ribosomes of bacteria) (Ex - tetracycline)
  2) DNA synthesis (Ex - Rifamycin)
  3) Break cell wall (made ~~of~~ of peptidoglycan)
  4)

→ Ex - Penicillin

→ Rifamycin ~~att~~ inhibits the action of RNA polymerase.

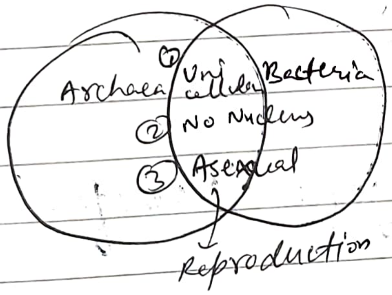
→

2 types of Bacteria

gram positive	gram negative
(thick cell wall)	(thin cell wall)
(gives purple colour) in exp	(~~gives~~ first remains colourless and then gives pink in safranin.

★ gram positive bacteria have a thick mesh-like cell wall made of peptidoglycan (50-90% of cell envelope) and as a result are stained purple by crystal violet, whereas gram-negative bacteria have a thinner layer (10% of cell envelope) so don't retain the purple stain and are counter-stained by safranin.

* Similarities between Archaea and Bacteria :-



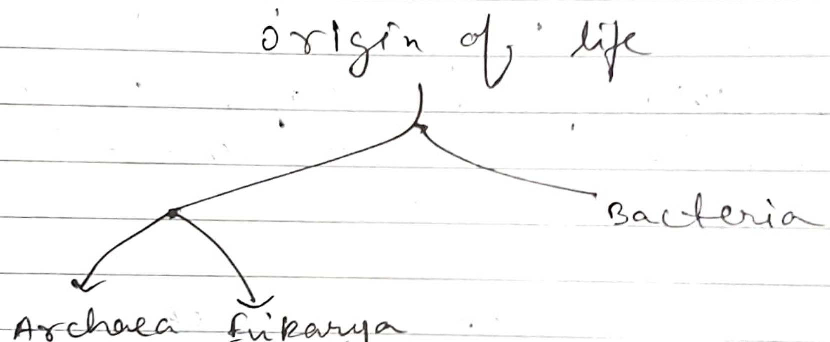
* Difference between Archaea and Bacteria :-

Archaea	Bacteria
1) Cell wall does NOT contain peptidoglycan.	1) Bacteria ~~cb~~ cell walls contain peptidoglycan.
2) Plasma membrane use isoprene chains	2) Plasma membrane use fatty acids chain.

NOTE The enzymes that read the genetic code in bacteria are different from the enzymes that read the genetic code in archaea.

NOTE Till date no archaea has caused any disease to humans.

* Archaeol is one of the main core ~~members~~ membrane lipids of archaea, one of the three domains of life.



(Both have same rRNA sequence)

(lateral gene transfer)

* A cell engulfs another cell through the process of endocytosis.

* Retroviruses take their genetic material, insert it into a host cell's genome, and then rely on the host to replicate that material.

NOTE HERV → Human Endogenous Retroviruses. (They make up 8% of our genetic code).

READ ABOUT HERV-H and HERV-K

* Types of diversity :-

1) Genetic diversity
2) Species diversity
3) Ecosystem diversity.

1) Genetic diversity is the variation in the genetic composition of individuals in a population, community or species.
→ Allows individuals to adapt to different conditions. Thus, high genetic diversity increases the ability of the populations and species to survive in the major changes in their environment such as climate change.

2) Species diversity - The richness of species in an ecosystem.

3) Ecosystem diversity describes the variation in all living and non-living things in a particular geographic or ecological Region.

NOTE → The known and described no. of species on earth is between 1.7 and 1.8 million.

→ Current estimates of the total no. of species range from 5 to 30 million.
→ About 61 %. of these species are insects.

Date
Page

* Threats to Biodiversity :-

1) Natural causes :-
- Narrow geographical Area
- Over population
- Low breeding rate
- Natural Disasters.

2) Anthropogenic (by Human) causes :-

- Pollution
- Hunting
- Global warming and climate change.
- Agriculture.

* Species Richness → No. of species per unit area.

* Species Abundance → Species abundance is the no. of individuals per ~~species~~ species in a community.

* Biodiversity is measured in 3 terms :-

Whittaker (1972) described three :-
1) Alpha diversity → refers to the diversity within a particular area or ecosystem, and is usually expressed by the number of species (i.e. - species richness) in that ecosystem.

2) Beta diversity : a comparison of diversity between ecosystems, usually measured as the amount of species change between the ecosystems.

) Gamma diversity :- a measure of the overall diversity within a large region. Geographic-scale species diversity according to Hunter (2002).

* Ecosystem services provided by biodiversity :-

1) Ecosystem stability
2) Nutrient Recycling
3) Energy transfer
4) Climate Stability
5) pollutant breakdown
6) Balance of Atmospheric gases.

* In recent years, detailed analysis of DNA sequences from a variety of Prokaryotic organisms revealed that they are of two distinct types: ~~so-called~~
1) so-called 'True' bacteria or Eubacteria
2) Archaea.

* In prokaryotes they have capsule. Its work is:-

1) firstly it is composed of polysaccharide (long chains of carbohydrate molecules).

2) The capsule which can be found in both gram negative and gram positive bacteria is different from the second lipid membrane - bacterial outer membrane which contains lipopolysaccharides and lipoproteins and is found only in gram negative

bacteria.

3) The capsule is considered a virulence factor because it enhances the ability of bacteria to cause disease (e.g - it prevents phagocytosis)

4) The capsule can protect cells from engulfment by eukaryotic cells, such as ~~mac~~ macrophages.

5) A capsule specific antibody may be required for phagocytosis to occur. Capsules also contain water which protects the bacteria against desiccation.

6) They also exclude bacterial viruses and most hydrophobic toxic materials such as detergents.

7) Capsules also help cells adhere to surfaces.

* Flagella :-

1) Flagella help move the bacteria toward nutrients and other attractants, a process called chemotaxis.

2) The long filament which acts as a propeller, is composed of many subunits of a single protein, flagellin, arranged in several intertwined chains.

3) The energy for movement, the proton motive force is provided by adenosine Triphosphate

pili :-

1) Pili are hair-like filaments that extend from the cell surface. They are shorter and straighter than flagella and are composed of subunits of pilin, a protein arranged in helical strands.

2) Pili are responsible for virulance in the pathogenic strains of many bacteria.

3) conjugative pili allows for the transfer of DNA between bacteria, in the process of bacterial conjugation. They are sometimes called "sex pili", in analogy to sexual reproduction, because they allow for the exchange of genes via the formation of "mating pairs".

4) pili are responsible for virulence in the pathogenic strains of many bacteria.

* Microfilaments and microtubules provide internal framework / ~~cyto~~ makes the cytoskeleton.

* Nucleolus functions as site of ribosome production, ribosomes then migrate to the cytoplasm through nuclear pores.

* Centrioles :- involved in the organization of mitotic spindle fibres during cell division and in the completion of cytokinesis.

* Centrosomes - serves as main microtubule organizing center (MTOC) of the animal cell, as well as a regulator of cell-cycle progression; are composed of two centrioles arranged at right angles to each other.

* In plant cells, the cell wall is made up of cellulose (polysaccharide), hemicellulose (polysaccharide), pectin (polysaccharide) and protein.

* In many fungi, the cell wall is formed of chitin (polysaccharide)

* In bacteria, the cell wall contains protein-lipid polysaccharide complexes.
Bacterial cell walls are made of peptidoglycan (also called murein), which is made ~~of~~ from polysaccharide chains cross-linked by peptides.

* The cell wall protect organism against viruses and other pathogens.

* pseudomurein is a type of cell wall found in some archaea, particularly ~~in members of the~~ order
Pseudomurein is resistant to lysozome.

NOTE * Plant cells have both cell wall and cell membrane.

* Both plant and animal cells have vacuoles. A plant cell contains a large single vacuole that is used for storage and maintaing the shape of the cell. In

contrast, animal cells have many, smaller vacuoles.

→ centriole is present only in Animals cells and in some lower plant cells. e.g. - mosses and ferns, but not present in higher plant cells.
Plants don't have centrioles but they possess microtubules which acts just like a centriole that is it helps in the spindle fibre (aggregates of microtubules that move chromosome during cell division) formation during cell division.

A centrosome comprises two microtubule rings known as centrioles. It's main function is to organise the microtubules and provide a structure to the cell.

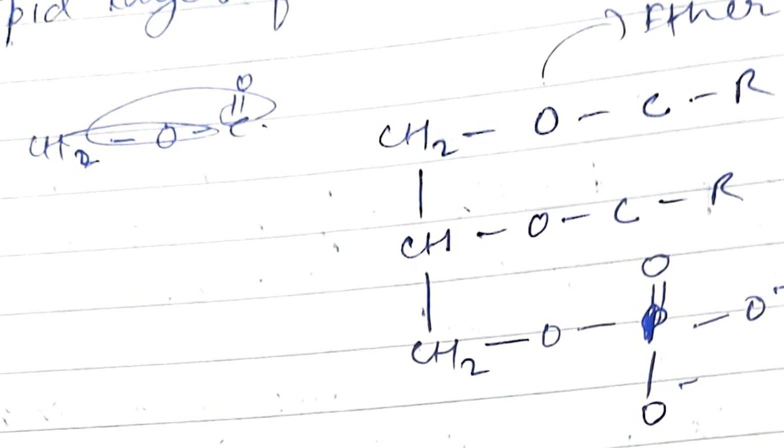
NOTE → All membranes contain two basic components: lipids and proteins. Some membranes also contain carbohydrate.

* Membrane lipids :-

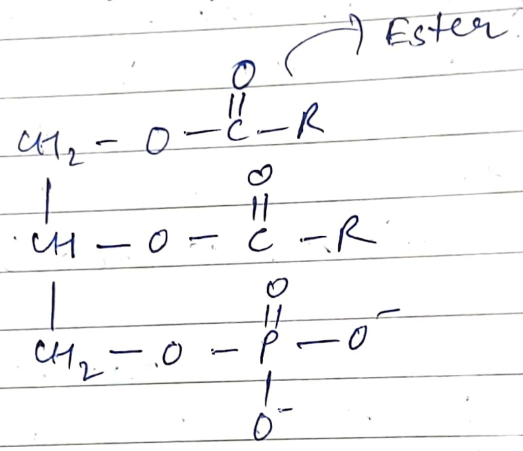
⇒ Lipids are water insoluble oily organic substance that are soluble in non aqueous solvents such as chloroform, ether etc.

⇒ In addition to being structural components of membranes, lipids serve as fuel molecules.

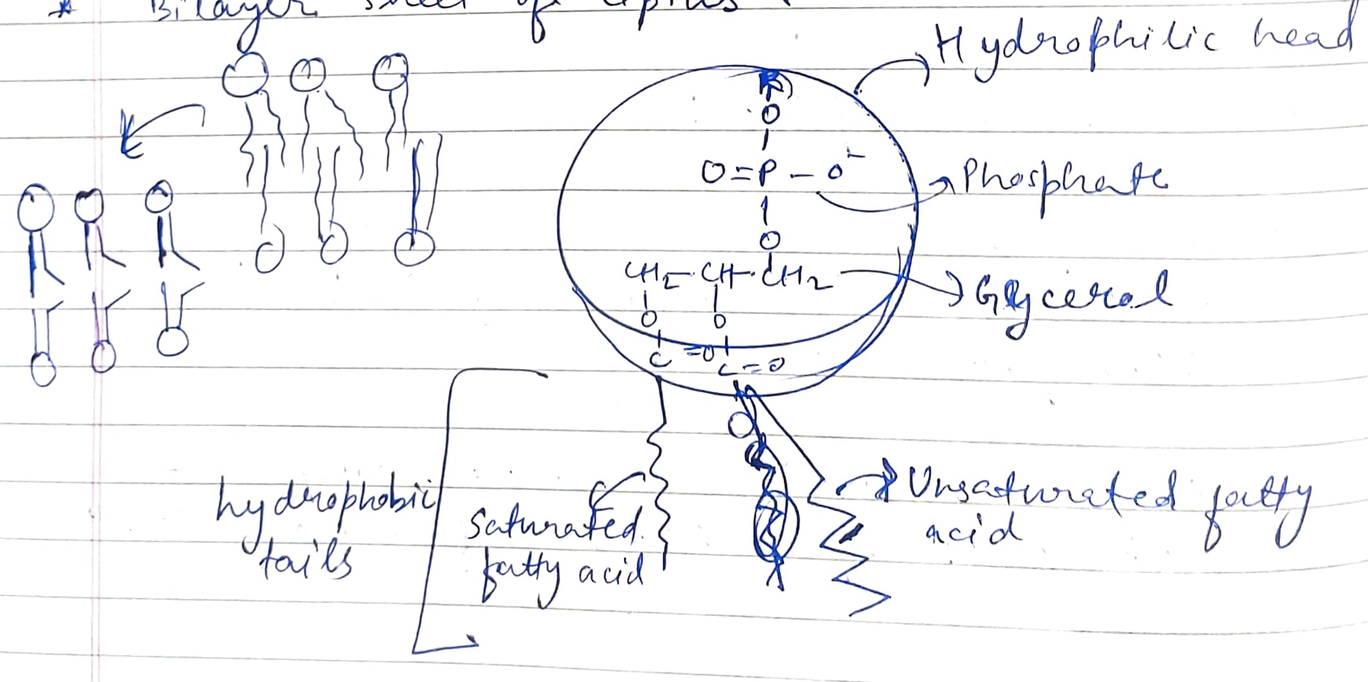
* Phospholipid layer from Archaea :-



* Phospholipid layer from Bacteria and Eukarya :-



* Bilayer sheet of lipids :-



* Saturated Fat	Unsaturated Fat
1) meats, butter, dairy products	1) vegetable oils
2) solid at room temp.	2) liquid at room temp.
3) increase levels of "bad cholesterol" (LDP)	3) increase levels of "good" cholesterol (HDP)
4) low-density lipoprotein clogs arteries	4) High-density lipoprotein, or HDL, "grabs" LDL and escorts it to the liver where LDL is broken down and eventually removed from the body

* Plasma membrane is semipermeable.

NOTE → cholesterols prevents phospholipids from (the bilayer) separating too far.

→ In fluid Mosaic Model, carbohydrates on the bilayer help in identification / recognition of own cell. (Blood type is determined by the carbohydrates on our cell).

→ Protein channels in the phospholipid bilayer allows certain objects to pass (Ex- glucose).

* Plasma membrane (or cell membrane) is also called fluid mosaic model.

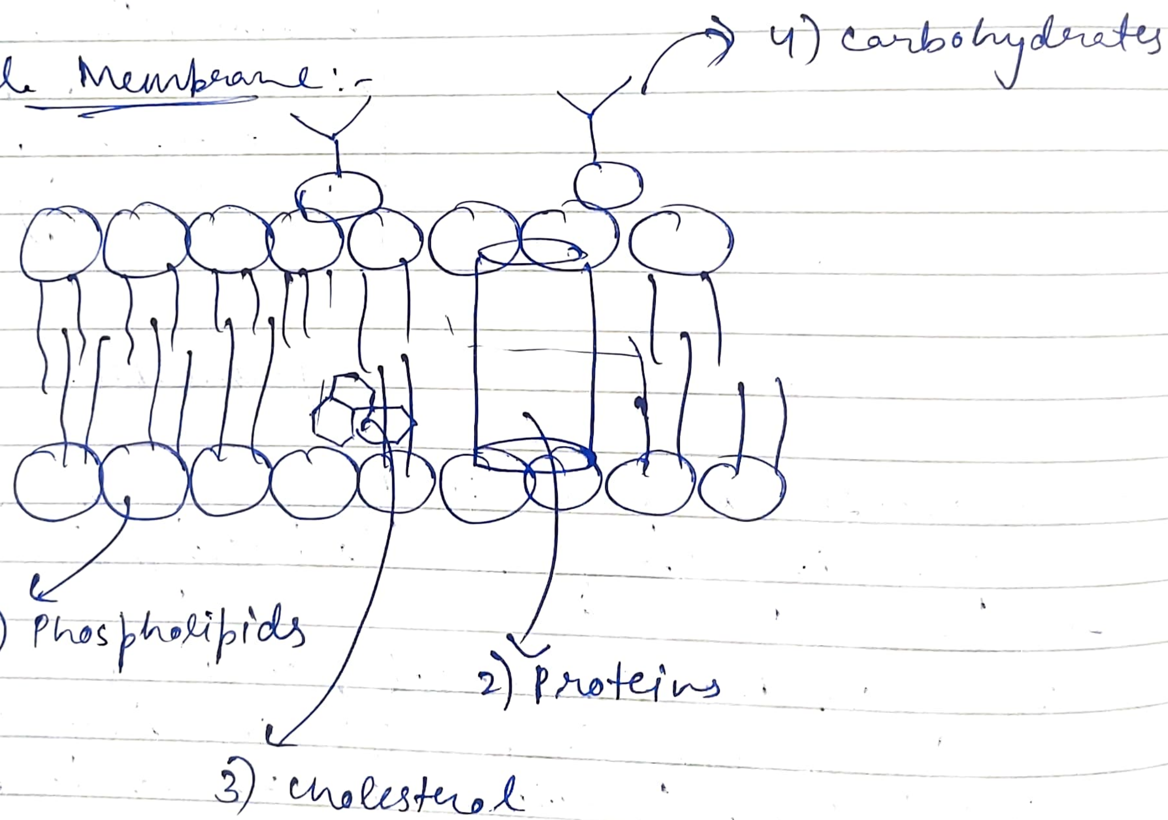
* Phospholipids consist of 3 main parts :-

1) Phosphate group
2) Glycerol Backbone
3) Fatty Acid Tails.

* Only Small and uncharged molecules pass easily through the phospholipid bilayer.

Imp

* Cell Membrane :-



NOTE * All cells have a flexible cell membrane.
* Most cells have a rigid cell wall.

* The cell organelles are of 3 types :-
i) Double membrane
ii) Single membrane
iii) No membrane.

i) Double membrane bound cell organelles :-
a) Nucleus
b) Mitochondria
c) Chloroplast

(ii) Single membrane bound cell organelles :-
a) Endoplasmic reticulum
b) Golgi apparatus
c) Vacuole
d) Lysosomes and endosomes.
e) Peroxisomes
~~f)~~

iii) Cell organelles without the membrane :-
a) Ribosomes
b) Centrosomes (Centriole)
c) Cytoskeleton (Microtubules, actin ~~or~~ filaments and intermediate filaments).

- **Nucleus** :-

→ Major part is made of chromosomes. Each chromosome is made up of DNA (2m long) wrapped around histone protein. (Histon octamer is composed of H2A, H2B, H3, and H4.)

* → Nuclear ~~porth~~ pore complexes :-

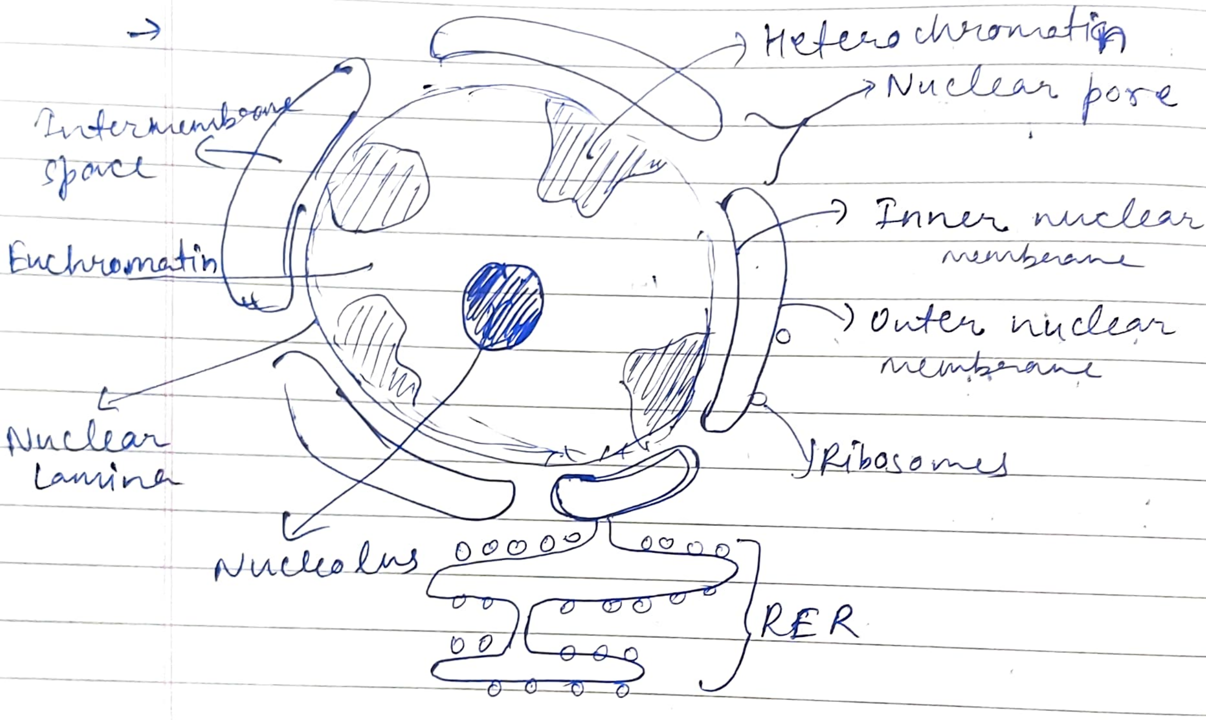
i) Nucleoporin

ii) Importins

iii) Exportins

→ READ FROM SLIDES (This part)

→



→ The 2 membranes fuse at nuclear pores.

→ Nuclear Lamina is composed of proteins called Lamins which gives the shape to nucleus.

→ Condensed DNA which are inactive for transcription are called Heterochromatin.

→ The light areas containing the loose chromatin (DNA and protein) structure (active for transcription are called Euchromatin.

→ Ribosomal RNA is synthesised in the Nucleolus.

NOTE → RBC does anaerobic respiration so doesn't contain ~~Ribosome~~ mitochondria.

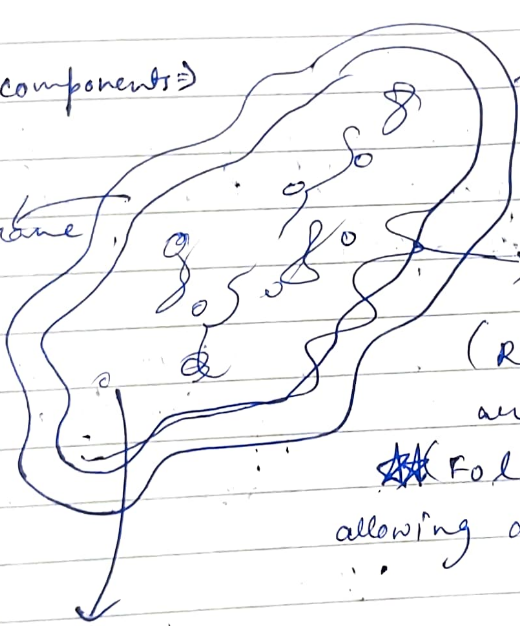
NOTE → Each human cell contains on avg 100-1000 of mitochondria.

MITOCHONDRIA :-

⇒ 4 main components ⇒

(1) Outer membrane (permeable to certain ions & small molecules)

(4) Intermembrane space (composition similar to cytosol)



(2) Inner membrane (Respiratory chain proteins are present)

Folded into multiple cristae allowing a large surface area)

(3) Matrix (Most of the metabolic rxns take place)

⇒ Function :-

i) Essential for aerobic metabolism

ii) Energy production through oxidative phosphorylation

iii) Involved in metabolic pathways :-

→ Beta oxidation

→ Krebs cycle

→ Synthesis of iron sulphur clusters.

4 bgic

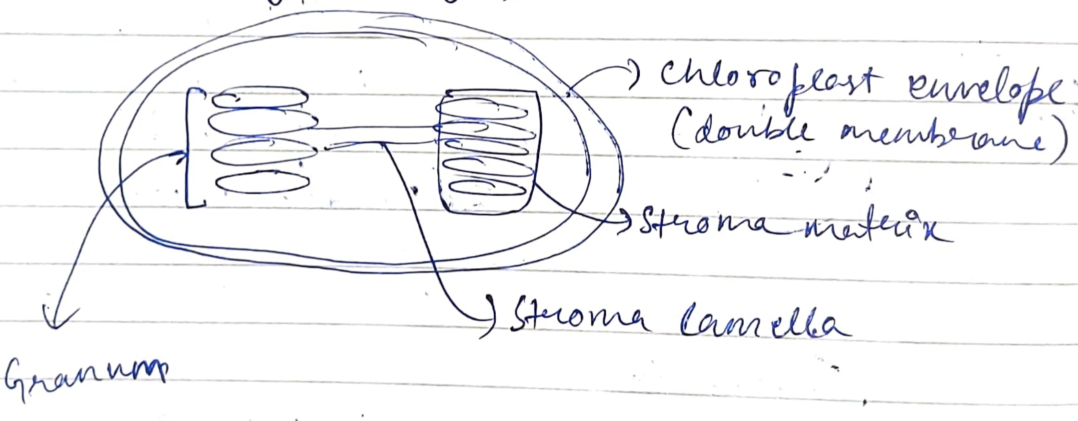
iv) Maintain, replicate & transcribe their own DNA.
v) Translate mRNA into protein
vi) Import and assembly of protein
vii) Production of reactive O_2 species

NOTE ⇒ ATP is generated through oxidative phosphorylation. (in the inner mitochondrial membrane).

■ CHLOROPLAST :-

⇒ Chloroplasts capture energy from sunlight in plant cells and in single-celled Eukaryotes such as Volvox (a green alga)

⇒



SINGLE MEMBRANE ORGANELLE

■ GOLGI APPARATUS -

■ Endoplasmic Reticulum →

⇒ The extensive network of closed, flattened membrane-bound sac like structures of ER are called Cisternae, whose interior is called Cisternal space or lumen.

⇒ ER is involved in Protein synthesis, Lipid metabolism and calcium storage.

Types of ER

SER
(without Ribosomes)
(synthesis of fatty acids and phospholipids)

RER
(with ribosomes)
(They synthesise membrane and organelle proteins and virtually all proteins required by the cell. ~~They~~
(They are involved in the synthesis, folding, modification and transport of proteins to rest of the cell).

■ GOLGI APPARATUS :-

⇒ It is made up of a series of flattened stacked pouches called Cisternae.

⇒ It is located in the cytoplasm next to the Endoplasmic Reticulum and near the Nucleus.

⇒ It helps in the processing and packaging of proteins and lipid molecules.

⇒ Proteins after synthesis in Rough ER leaves the organelle within a small membrane bound vesicles (which originate from Rough ER)

→ These vesicles act as transport vesicles which carry the proteins to Golgi complex.

→ Str. Golgi Apparatus :-

Trans Golgi Network

NOTE → CIS (means face towards ER)